MZ2POL-08: Templated Policies for MZ Migration

> **Series:** MZ2POL — Management Zone to Policy Migration | **Notebook:** 9 of 14 | **Created:** February 2026 | **Last Updated:** 10/13/2026

Overview

When migrating from Management Zones at scale, organizations with dozens of MZ-based teams face the challenge of creating and maintaining an equivalent number of IAM policies. Policy templating with `${bindParam:...}` parameters eliminates this duplication.

This notebook shows how to convert common MZ access patterns into parameterized policy templates, bind them at scale via the IAM API, and validate the migrated access using DQL queries.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM and Grail enabled
Permissions	Account permission **View and manage users and groups** (`account-user-management`) for group and binding management — there is no `account-iam-admin` permission (Role-based permissions (DT docs))
API Access	OAuth client or API token with IAM management scopes
Prior Knowledge	**MZ2POL-04: Policies and Boundaries** (G-P-B pattern, policy syntax)
Migration Status	MZ assessment complete (**MZ2POL-03**), security context assigned

1. Why Templates Matter for MZ Migration

The Scale Problem

The average Dynatrace customer has **92 Management Zones** with **1,700 conditions**. Each MZ needs a policy equivalent. Without templates, that means creating and maintaining 92 individual policies.

Approach	Policies to Create	Maintenance Effort	Audit Complexity
One policy per MZ	92	High — each permission change requires 92 edits	92 policies to review
Templated policies	3-5 templates	Low — edit template once, all bindings updated	3-5 templates + bindings

Where Templates Fit in the 6-Phase Migration

Templates are most impactful in Phases 3-5 of the migration (from **MZ2POL-06**):

Phase	Activity	Template Role
Phase 1: Assessment	Inventory MZs	Classify MZ types → identify which template to use
Phase 2: Tagging	Set security context	Values become template parameter values
Phase 3: Access Management	Create policies	**Create 3-5 templates instead of 92 policies**
Phase 4: Implementation	Bind to groups	**Bulk bind via IAM API script**
Phase 5: Validation	Verify access	**DQL queries to validate bindings**
Phase 6: Cutover	Remove MZs	No template impact

2. Converting MZ Patterns to Parameterized Policies

Each MZ type (from the **MZ2POL REFERENCE**) maps to a specific template pattern.

Vertical MZ (Environment-Based)

Before: MZ "Production" with host group rules.

Template:

```
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,
      storage:events:read, storage:bizevents:read
  WHERE storage:dt.security_context = "${bindParam:environment}";
ALLOW settings:objects:read WHERE settings:dt.security_context =
"${bindParam:environment}";
```

Bindings:
- Production team: `{"parameters": {"environment": "prod"}}`
- Staging team: `{"parameters": {"environment": "staging"}}`

Horizontal MZ (Team-Based)

Before: MZ "Frontend-Team" with service tag rules.

Template:

```
ALLOW storage:logs:read WHERE storage:dt.security_context =
"${bindParam:team}";
ALLOW storage:spans:read WHERE storage:dt.security_context =
"${bindParam:team}";
ALLOW storage:metrics:read WHERE storage:dt.security_context =
"${bindParam:team}";
```

Binding: `{"parameters": {"team": "team-frontend"}}`

LOB MZ (Line of Business)

Before: MZ "LOB5" with combined host group + service tag rules.

Template (three-domain, matching MZ2POL-04 best practice):

```
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,
      storage:events:read, storage:bizevents:read
  WHERE storage:dt.security_context = "${bindParam:lob}";
ALLOW settings:objects:read, settings:objects:write
  WHERE settings:dt.security_context = "${bindParam:lob}";
```

> **Storage writes cannot be condition-scoped (live-verified 07/2026):** wildcards are rejected and `storage:logs:write WHERE ...` fails with `Invalid condition name`. Where a LOB genuinely needs ingest/write access, grant `storage:logs:write, storage:metrics:write, storage:events:write` unscoped in a separate policy.
>
> ⚠️ **`storage:events:read` with `=` or `startsWith` may match none of a team's events — test before rollout.** Grail's permission docs: *"Using `=`, `STARTSWITH` or `IN` when the field holds an array will always return `false`. If you expect your record filters might contain an array, use the `MATCH` operator in your IAM statements."* On the validation tenant (10/05/2026, 24 h), **every** `events` record that carried `dt.security_context` carried it as an array — `DAVIS_EVENT` 72,235, `SYNTHETIC_EVENT` 27,032, `FLEET_EVENT` 4,576, `DAVIS_PROBLEM` 460, no string-typed record — while logs carried it as a string. A template that bundles `storage:events:read` into a `= "${bindParam:…}"` or `startsWith` statement can therefore give the bound group none of its own Davis events, problems or synthetic events, and nothing errors. The policy reference lists `match` as an operator for `storage:dt.security_context` on `storage:events:read`, but the templating docs show bound parameters only with `=` and `IN`, so whether `MATCH`

> `("${bindParam:team}")` is accepted is not documented. Before rolling these templates out, bind one to a test group and confirm a member can `fetch events` for the team's context; if not, grant events in a separate, non-templated statement that uses `MATCH`.
>
> **Sources:** Permissions in Grail (DT docs) — *"Using `=`, `STARTSWITH` or `IN` when the field holds an array will always return `false`."*, IAM policy templating (DT docs), IAM policy reference (DT docs).

Combined with **two parallel boundaries** using the same scope (Gen2 boundary attaches to Gen2 policy, Gen3 boundary attaches to Gen3 policy):

```
// Gen3 boundary — pairs with Gen3 policies (storage:* / settings:*)
storage:dt.security_context IN ("LOB5");
settings:dt.security_context IN ("LOB5");
```

```
// Gen2 boundary — pairs with Gen2 policies (environment:*), transitional
environment:management-zone IN ("LOB5");
```

Binding: `{"parameters": {"lob": "LOB5"}}` — applied to both bindings on the group.

Conversion Decision Table

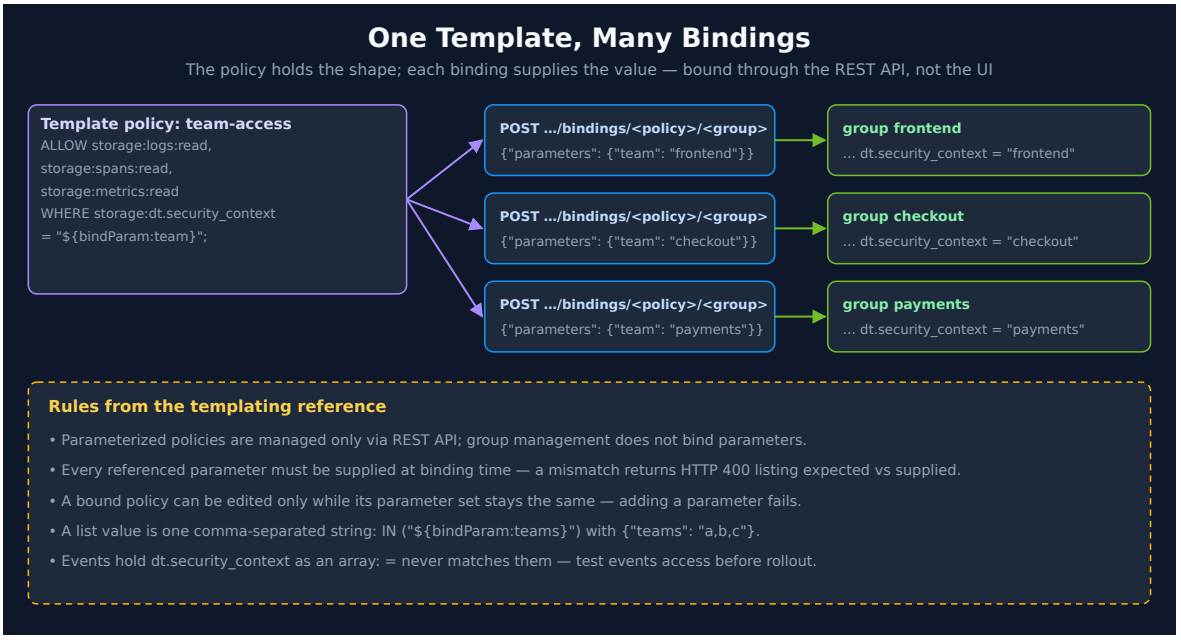
MZ Type	Template Type	Parameters	Boundary Needed?
Vertical (environment)	Environment-scoped	`environment`	Yes — restrict to environment
Horizontal (team)	Team-scoped	`team`	Yes — restrict to security context
LOB	Full three-domain	`lob`	Yes — three-domain boundary

| Bucket-based | Bucket-scoped | `log-bucket`, `span-bucket` | Optional |

> **Transversal access tip:** For organisations where some teams need cross-application access by component type (database, networking, OS), consider a structured `comp:<component>/bu:<business-unit>/app:<application>` context format. A single template with `MATCH('comp:db*')` then covers all database-layer

> data across every application — no per-app policy per team needed. See **IAM-04: Policy Authoring** for the full pattern.

How a Template Becomes Many Policies



> **Sources:** Policy templating (DT docs) — *"Parameterized policies can be managed only via REST API."*, *"Currently, group management does not support binding policies with parameters."* and *"Modification of bound policies is allowed only if the parameter set doesn't change"*

3. Template Patterns for Common MZ Replacements

These patterns map directly from MZ permission levels to template bindings.

Pattern A: Read-Only Team Access

Replaces: User assigned "Viewer" role in a team MZ.

Template: `tpl-team-data-reader`

```
ALLOW storage:logs:read WHERE storage:dt.security_context =
"${bindParam:team}";
ALLOW storage:spans:read WHERE storage:dt.security_context =
"${bindParam:team}";
ALLOW storage:metrics:read WHERE storage:dt.security_context =
"${bindParam:team}";
ALLOW document:documents:read;
```

Binding: `{"parameters": {"team": "checkout"}}`

Pattern B: Full Team Access

Replaces: User assigned "Editor" role in a team MZ.

Template: `tpl-team-data-editor`

```
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,
      storage:events:read, storage:bizevents:read
  WHERE storage:dt.security_context = "${bindParam:team}";
ALLOW settings:objects:read, settings:objects:write
  WHERE settings:dt.security_context = "${bindParam:team}";
ALLOW document:documents:read, document:documents:write,
document:documents:delete;
```

Binding: `{"parameters": {"team": "checkout"}}`

Pattern C: Multi-MZ Access

Replaces: User assigned to multiple MZs (e.g., "Checkout" AND "Shared-Services").

Solution: Create **two bindings** of the same template with different parameter values to the same group:

```
// Binding 1: Checkout access
curl -X POST "${API_BASE}/bindings/${POLICY_UUID}/${GROUP_UUID}" \
  -d '{"parameters": {"team": "checkout"}}'

// Binding 2: Shared services access
curl -X POST "${API_BASE}/bindings/${POLICY_UUID}/${GROUP_UUID}" \
  -d '{"parameters": {"team": "shared-services"}}'
```

Both bindings are additive — the user gets access to both scopes.

Pattern D: Regional MZ Replacement

Replaces: MZ "US-East" with region-based host group.

Template: `tpl-region-reader`

```
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,
      storage:events:read, storage:bizevents:read
  WHERE storage:dt.security_context startsWith "${bindParam:region-prefix}";
ALLOW settings:objects:read WHERE settings:dt.security_context startsWith
"${bindParam:region-prefix}";
```

Binding: `{"parameters": {"region-prefix": "us-east"}}`

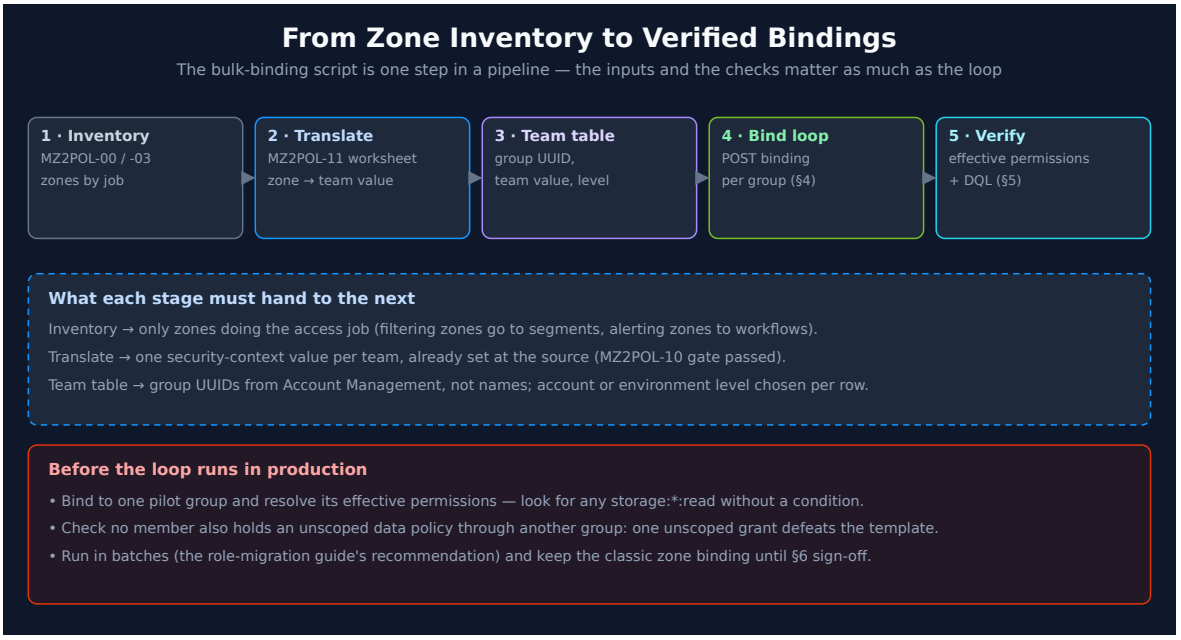
> **Patterns B and D grant `storage:events:read` with `=` / `startsWith`.** Event records hold `dt.security_context` as an array, which those operators never match — read the warning in §2 and test a binding before rollout.

Pattern Summary

Pattern	MZ Permission Level	Template	Key Difference
A	Viewer in MZ	`tpl-team-data-reader`	Read only
B	Editor in MZ	`tpl-team-data-editor`	Read + write
C	Multiple MZs	Same template, multiple bindings	Additive access
D	Regional MZ	`tpl-region-reader`	Prefix match

4. Bulk Binding via IAM API

Organizations with 92 MZs cannot bind one at a time via the UI. Use the IAM API for bulk operations — templated bindings cannot be made in the UI at all.



> **Sources:** Upgrade role-based permissions to Dynatrace IAM policies (DT docs) — *"Consider deploying your changes in batches rather than all at once."*

Preparation Checklist

Before running bulk binding:

1. **Export MZ list** — Use **MZ2POL-00** SDK tool or Settings API
2. **Map each MZ to a group UUID** — List groups via `GET /iam/v1/accounts/{ACCOUNT}/groups`
3. **Map each MZ to a security context value** — This becomes the parameter value
4. **Choose the appropriate template** — Reader, editor, or three-domain

Bulk Binding Script

```
#!/bin/bash
# Bulk bind team-access template to all team groups
ACCOUNT_UUID="&lt;your-account-uuid&gt;"
POLICY_UUID="&lt;team-reader-template-uuid&gt;"
TOKEN="&lt;bearer-token&gt;"
API_BASE="https://api.dynatrace.com/iam/v1/repo/account/${ACCOUNT_UUID}"

# Map: team name → group UUID
declare -A TEAM_GROUPS=(
  ["checkout"]="&lt;group-uuid-checkout&gt;"
  ["payments"]="&lt;group-uuid-payments&gt;"
  ["catalog"]="&lt;group-uuid-catalog&gt;"
  ["search"]="&lt;group-uuid-search&gt;"
  ["frontend"]="&lt;group-uuid-frontend&gt;"
)

echo "Binding template ${POLICY_UUID} to ${#TEAM_GROUPS[@]} groups..."
for team in "${!TEAM_GROUPS[@]}"; do
  GROUP_UUID="${TEAM_GROUPS[$team]}"
  echo -n "  ${team} (${GROUP_UUID})... "
  HTTP_CODE=$(curl -s -X POST \
    "${API_BASE}/bindings/${POLICY_UUID}/${GROUP_UUID}" \
    -H "Authorization: Bearer ${TOKEN}" \
    -H 'Content-Type: application/json' \
    -d "{\"parameters\": {\"team\": \"${team}\"}}" \
    -o /dev/null -w "%{http_code}")
  if [ "$HTTP_CODE" == "201" ]; then
    echo "Created"
  elif [ "$HTTP_CODE" == "409" ]; then
    echo "Already exists"
  else
    echo "ERROR (HTTP ${HTTP_CODE})"
  fi
done
echo "Done."
```

Verification Script

After bulk binding, verify all bindings exist:

```
# List all bindings for the template policy
curl -s -X GET \
  "${API_BASE}/bindings/${POLICY_UUID}" \
  -H "Authorization: Bearer ${TOKEN}" | python3 -m json.tool
```

Error Handling

HTTP Code	Meaning	Action
201	Binding created	Success
400	Bad request	Check parameter names match policy template
401	Unauthorized	Refresh token
404	Policy/group not found	Verify UUIDs
409	Binding already exists	Skip or update

5. Validating Migrated Access with DQL

After binding templated policies, validate that migrated access matches the original MZ-based access.

```
// Validate: All services have security context for migration
fetch dt.entity.service
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missingContext = countIf(isNull(dt.security_context))
| fields total, withContext, missingContext,
         coveragePercent = round(100.0 * withContext / total, decimals: 2)

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

```
// Compare: MZ membership vs security context value alignment
// Replace "Frontend-Team" and "team-frontend" with your MZ and context values
fetch dt.entity.service
| summarize
    inMZ = countIf(in(managementZones, {"Frontend-Team"})),
    hasContext = countIf(matchesValue(dt.security_context, "team-frontend")),
    inBoth = countIf(in(managementZones, {"Frontend-Team"}) and
matchesValue(dt.security_context, "team-frontend"))
| fields inMZ, hasContext, inBoth,
         match = if(inMZ == inBoth, then: "ALIGNED", else: "GAPS EXIST")

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query —
it inspects
// the classic constructs this migration replaces, so keep the classic query
above:
//   - management zones have NO Smartscape node equivalent; they are what this
migration
//     replaces with segments and policies.
//   - dt.security_context is an ARRAY on dt.entity.service too, so it is
tested with
//     matchesValue (matches any element). dt.security_context == "team-
frontend" compares an
//     array to a string, is always false, and made every run report "GAPS
EXIST".
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when
unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a
Smartscape rewrite
//     would miscount coverage.
```

```
// Find entities in MZ but without matching security context
// These need security context before template binding will work
// dt.security_context is an ARRAY here too: `!= "team-frontend"` is always
true (Grail INFO
// EQUALITY_COMPARISON_OF_INCOMPATIBLE_TYPES) and would list every MZ member.
Use not in(...).
fetch dt.entity.service
| filter in(managementZones, {"Frontend-Team"})
| filter not in("team-frontend", dt.security_context) or
isNull(dt.security_context)
| fields entity.name, dt.security_context, managementZones, tags
| sort entity.name asc
| limit 50

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query —
it inspects
// the classic constructs this migration replaces, so keep the classic query
above:
//   - management zones have NO Smartscape node equivalent; they are what this
migration
//     replaces with segments and policies.
//   - dt.security_context is an ARRAY on classic entities as well, but null
when unset;
//     on Smartscape nodes it is an empty [] when unset, not null, so
//     isNull / isNotNull(dt.security_context) does not carry over — a
Smartscape rewrite
//     would miscount coverage.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via
getNodeField).
```

```
// IAM API activity against this environment (migration tracking aid, NOT a
change log)
// Data object corrected 09/24/2026. The Dynatrace audit trail is NOT in
`logs`: the former
// `fetch logs | filter matchesPhrase(log.source, "audit")` matched nothing,
or matched an
// unrelated file-based audit log (a database .aud file on the validation
tenant). Environment
// audit records are structured events in `dt.system.events` with event.kind
== "AUDIT_EVENT".
// Group, policy, boundary and binding changes are ACCOUNT-level. They are not
in an
// environment's audit events; review them in Account Management &gt; Settings
&gt; Audit log.
// Policy bindings are account-level, so track binding changes in that audit
log. This query
// shows IAM API calls against the environment - useful for spotting
automation. Scheduled
// lookup-file uploads (/lookups/iam_*) are excluded; they are not IAM
changes.
fetch dt.system.events, from:-30d
| filter event.kind == "AUDIT_EVENT"
| filter contains(resource, "iam") and not startsWith(resource, "/lookups/")
| summarize {requests = count(), last_seen = takeMax(timestamp)}, by:
{event.type, user.id}
| sort requests desc
| limit 100
```

6. Before/After Comparison Patterns

Validation Framework

For each migrated MZ, verify access parity:

Validation	MZ-Based (Before)	Template-Based (After)	Status
User can query logs in scope	Via MZ filter	Via security context + template binding	☐ Test
User cannot query logs out of scope	MZ excludes	Boundary restricts	☐ Test

Validation	MZ-Based (Before)	Template-Based (After)	Status
Settings access scoped	MZ-based	Settings security context in template	☐ Test
Dashboard access	MZ filter	Segment + boundary	☐ Test

Parallel Running Strategy

During validation, keep both access models active:

1. **Keep MZ active** — Don't remove MZs until template-based access is validated
2. **Add template binding** — Users gain access via both paths temporarily
3. **Test with each model** — Verify same data visible via MZ and via template
4. **Remove MZ assignment** — Only after confirming parity
5. **Monitor for issues** — Watch audit logs for denied events post-cutover

Migration Completion Checklist (Per MZ)

- [ ] Security context set on all entities that were in this MZ
- [ ] Template policy created (or existing template identified)
- [ ] Binding created with correct parameter value
- [ ] Boundary created with matching scope (all three domains)
- [ ] DQL validation — security context coverage matches MZ membership
- [ ] User testing — access parity confirmed
- [ ] MZ assignment removed from RBAC
- [ ] Audit log reviewed for denied events

For full validation procedures, see **MZ2POL-07: Validation and Troubleshooting**.

For comprehensive template syntax and patterns, see **IAM-10: Templated Policy-Group Assignments**.

Summary

In this notebook, you learned:

1. Why policy templates are essential for large-scale MZ migration (3-5 templates vs 92 policies)
2. How to convert vertical, horizontal, and LOB MZ patterns to parameterized policies
3. Four template patterns for common MZ replacement scenarios
4. Bulk binding automation via the IAM API
5. DQL queries to validate that migrated access matches original MZ access
6. Before/after comparison patterns for migration verification

Next Steps

- **MZ2POL-07: Validation and Troubleshooting** — Full validation procedures and ongoing maintenance
- **IAM-10: Templated Policy-Group Assignments** — Deep dive on template syntax, Monaco automation, and advanced patterns

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References

- Policy Templating
- Policy Management API
- Working with Policies
- Access Management Concepts

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